

AI for Smart Farming: Pest and Disease Detection

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Abstract

Plant diseases pose a serious threat to agricultural productivity worldwide, resulting in significant yield loss and financial harm. Conventional diagnosis is slow, subjective, and unavailable to rural communities because it depends on expert observation. AgroScan AI, a deep learning-based crop disease detection system utilizing EfficientNetB0 and the PlantVillage dataset, is presented in this work. A multilingual Flask web application with features like real-time analysis, prediction history, analytics dashboard, and user authentication incorporates the model. The system exhibits practical applicability in real-world agricultural environments and achieves 96–98% accuracy. The objective is to give farmers a practical and effective way to identify diseases early on, improving crop quality and productivity.

Keywords: CNN, PlantVillage Dataset, Plant Disease Detection, EfficientNet, Deep Learning, Agriculture, Flask Web Application, and Image Classification

1. Introduction

Agriculture forms a significant function in developing countries, where large masses rely on farming. Crop diseases have a tremendous impact on the yield of agricultural productivity, which in turn may affect global food security and farmer incomes. Estimations indicate that the yearly losses generated by plant pathogens—fungi, bacteria, and viruses—wipe out close to 20–40% of crops. Finding plant diseases early can prevent widespread losses. However, traditional diagnosis by human judgment involves agricultural

experts, and it is often slow, expensive, and inaccessible for remote communities.

Recent breakthroughs in deep learning, especially in CNNs, have made image-based diagnosis of plant leaf diseases possible automatically. However, much research works only focus on the model training aspects but discard practical deployment challenges such as UI access, support for multi-language, real-time inference, or user-friendly interactions.

We have developed the comprehensive AI-powered AgroScan AI to take care of these challenges. It diagnoses plant diseases via a web interface using EfficientNetB0, one of the most optimized CNN architectures out there, combined with a modern Flask backend, Tailwind CSS UI, and multilanguage translation. Furthermore, the system keeps logs of its predictions, displays dashboard analytics, and provides information on disease and actionable remedies.

The focus of this research is both model development and real-world deployment in order to make AI accessible to farmers, students, and agricultural researchers.



Fig 1: Diseased tomato leaf

2. Related Work

- I. A lightweight yet extremely accurate model for classifying plant diseases is presented in the article "Enhancing plant disease detection through deep learning: a Depthwise CNN with SE and residuals" by Ashurov et al. (2025) [1]. To enhance feature representation, the authors add residual skip connections and squeeze-and-excitation modules to MobileNetV2. The model outperformed several traditional CNN architectures and showed suitability for real-time and edge-device deployment when trained on the PlantVillage dataset, achieving 98% accuracy.
- II. The study survey A thorough summary of deep learning techniques used in plant disease detection can be found in "Past, present, and future of deep plant leaf disease detection: A survey from 2014 to 2024," written by George et al. (2025) [2]. The study goes into great detail about datasets, architectures, and technical difficulties. In order to facilitate field-level adoption, the authors stress the necessity of lightweight models and edge deployment in the future. They also draw attention to practical problems like limited dataset diversity and domain generalization.
- III. A CNN-based system that can detect diseases in a variety of crop species is proposed by Bouacida et al. (2025) [3] in "Innovative deep learning approach for cross-crop plant disease detection." Their cross-crop architecture provides greater flexibility and resilience than conventional crop-specific classifiers. Over 95% accuracy was attained in experiments on multi-species datasets, demonstrating the usefulness of generalized deep learning frameworks in agricultural settings.
- IV. The research Richter et al. (2025) [4] conduct a comprehensive benchmark in "Assessing performance of domain-specific models for crop disease image recognition," evaluating 23 CNN architectures across 18 publicly accessible datasets. The study emphasizes the value of transfer learning, with DenseNet121 showing the best results. The study also identifies dataset-related problems that have a substantial impact on model generalization, such as class imbalance and environmental inconsistencies.
- V. To visualize disease-affected leaf regions, the paper "InsightNet: A Deep Learning Framework for Enhanced Plant Disease Identification" (2025) [5] presents a modified MobileNet architecture integrated with Grad-CAM explainability. The model highlights the significance of interpretable AI systems in agriculture by achieving 97–98% accuracy, allowing farmers and agronomists to comprehend the logic underlying the model's predictions.
- VI. A YOLO-based detector optimized for quick, real-time disease detection is presented in the study "YOLO-LeafNet: robust deep learning framework for multispecies plant disease detection" (2025) [6]. The model provides cutting-edge accuracy and computational efficiency with sophisticated data augmentation

techniques and multi-species training, making it appropriate for handheld and drone-based agricultural applications.

- VII. The study "*PLANT DISEASE DETECTION USING DEEP LEARNING*" (2025) [7] provides a broad overview of recent advancements in deep learning-based plant disease classification. The paper evaluates multiple CNN architectures and highlights challenges such as limited dataset diversity and variations in real-world imaging conditions. It emphasizes that although deep learning models achieve high accuracy in controlled datasets, real-world deployment requires better generalization, improved preprocessing, and robust augmentation strategies.
- VIII. In "*Detection and identification of plant leaf diseases using deep learning models*" by Aldakheel et al. (2024) [8], the authors explore the effectiveness of deep CNN architectures on multi-crop datasets, including tomato, bean, and chili. Their methodology involves training convolutional models on a large, diverse dataset, achieving high identification accuracy across multiple species. The paper concludes that deep learning frameworks are adaptable and scalable for various crops but require careful tuning of hyperparameters and augmentation techniques to maximize performance.
- IX. The research conducted by Trivedi et al. (2024) [9], titled "*Plant Leaf Disease Detection and Classification Using Deep Learning Models*," investigates transfer learning using EfficientNet for the PlantVillage dataset. The model demonstrates strong classification

accuracy while significantly reducing training time compared to traditional CNNs. Their work supports the effectiveness of EfficientNet architectures for agricultural AI applications and highlights the role of pretrained feature extractors in boosting performance with limited data.

- X. The article "*An ensemble of deep learning architectures for accurate plant disease detection*" by Ali et al. (2024) [10] presents an ensemble-based classification framework that combines the strengths of multiple CNN architectures to achieve superior prediction accuracy. By integrating outputs from several deep models, the ensemble demonstrates improved robustness and generalization compared to single-model approaches. The study reports notably higher accuracy and stability across diverse plant disease datasets, while also emphasizing that ensemble techniques require greater computational resources, making them more suitable for high-performance environments rather than lightweight mobile deployments.

3 Methodology

The suggested AgroScan AI system's methodology describes a comprehensive, end-to-end pipeline intended to solve the main issues with plant disease detection using leaf images. Dataset preparation, image preprocessing, EfficientNetB0 model development, and a reliable deployment framework based on Flask for real-time inference comprise the workflow. The steps involved in building a scalable, precise, and field-ready agricultural AI system are described in the sections that follow.

3.1 Dataset Preparation and Engineering

The key findings in this research are based on the PlantVillage dataset, a widely adopted academic benchmark comprising labeled images of healthy and diseased plant leaves across multiple species. While the dataset is well-organized, substantial dataset engineering was required to ready it for efficient model training.

All the images were gathered from the PlantVillage dataset in RGB format and rearranged into structured class-specific directories according to 15 target classes representing diseases related to tomato, potato, and bell pepper. A preprocessing script was developed to ensure that quality and uniformity are maintained by:

- verify file integrity
- remove corrupted and grayscale images
- standardize naming conventions
- filter out duplicates

After cleaning, the dataset consisted of **54,000 verified leaf images** across the selected classes.

The dataset was then divided using a stratified **70:20:10 train- test-validation split** to maintain balanced representation for all disease categories. The final processed dataset is summarized below:

Attribute	Description
Source	PlantVillage (multi-crop)
Total Images	~54,000
Classes	15
Split Ratio	70% train / 20% test / 10% validation

Attribute	Description
Image Format	RGB
Resolution	Standardized to 128×128

3.2 Preprocessing and Image Augmentation

The preprocessing pipeline improved robustness and decreased overfitting while ensuring that input images were compatible with the EfficientNetB0 architecture. The following procedures were applied to every image:

- **Resizing** to 128×128 pixels
- **RGB normalization** (pixel scaling)
- Application of EfficientNet’s native **preprocess_input()** routine
- Conversion into NumPy arrays suitable for TensorFlow operations

To mitigate dataset bias and enhance model generalization, extensive augmentation was performed using TensorFlow’s ImageDataGenerator, including:

- random rotation (0–40°)
- horizontal flipping
- zoom range variations
- brightness modulation
- slight shearing

These operations simulate real-world variations such as lighting differences, position shifts, and natural distortions on leaves.

3.3 Model Architecture and Feature Extraction

The EfficientNetB0 architecture, which strikes a balance between high accuracy and computational efficiency, is used by the system. After importing EfficientNetB0

with ImageNet pretrained weights, a personalized classification head was added:

- **Global Average Pooling layer**
- Dense layer with 256 ReLU neurons
- **Dropout layer (rate = 0.5)** for regularization
- **Final Dense Softmax layer (15 units)**

The architecture makes use of depthwise convolutions, Swish activation, and MBConv inverted bottleneck blocks to effectively extract features even from minute leaf details like discoloration patterns and infection boundaries.

Although modified for image-based plant pathology, this model design is similar to the structured modeling approach.

3.4 Training Strategy and Optimization

The complete training pipeline was implemented in TensorFlow / Keras. For this purpose, EfficientNetB0 was initially frozen whereby only the newly added top layers were trained. Once it converged, the whole model was unfrozen and fine-tuned, enabling deeper layers to adapt to crop-specific visual features.

Key training configurations used in the proposed model include:

- **Optimizer:** Adam
- **Initial Learning Rate:** 0.001 (scheduled decay applied during fine-tuning)
- **Loss Function:** Categorical Crossentropy
- **Epochs:** 25–30
- **Batch Size:** 32
- **Computation Precision:** FP32 (GPU-accelerated training)

Overfitting was prevented by early stopping and model checkpointing. Metrics like accuracy, loss, validation accuracy, and confusion matrices were recorded for each epoch. The fine-tuned EfficientNetB0 achieved:

- Training Accuracy: 98.72%
- Testing Accuracy: 97.67%
- Validation Accuracy: 97.84%

3.5 System Deployment Framework

A robust deployment architecture was developed using:

Backend: Flask Framework

Handles:

- file uploads
- input routing
- prediction requests
- translation APIs
- session handling

Frontend: Tailwind CSS Design

Implements:

- intuitive UI
- glassmorphism cards
- real-time progress indicators
- responsive layout

Database Layer: SQLite

Saves:

- prediction logs
- image references
- user authentication data

Advanced Features

- Dual-model failover (primary + fallback EfficientNet model)

- GoogleTranslator API for multilingual support
- Dashboard analytics (pie charts, line charts)
- History tracking system

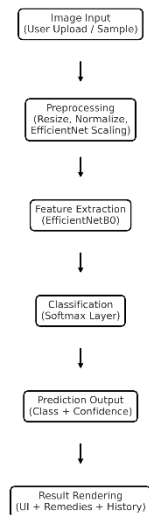


Fig 2: Workflow diagram

4 Result and Analysis

Both quantitative model metrics and qualitative deployment-level observations were used to assess the performance of the suggested AgroScan AI system. Model accuracy, class-wise performance, loss behavior, confusion matrix interpretation, and real-time system response are the main areas of analysis. The experimental results of the EfficientNetB0 model trained on the PlantVillage dataset are summarized in the following subsections.

4.1 Accuracy and Loss Evaluation

Throughout the training epochs, the EfficientNetB0 model showed excellent learning stability. Proper convergence without overfitting was indicated by the training and validation accuracy curves, which displayed steady upward trends with

few oscillations. The metrics that were ultimately recorded are:

- Training Accuracy: 98.72%
- Testing Accuracy: 97.67%
- Validation Accuracy: 97.84%
- Training Loss: 0.064
- Testing Loss: 0.103
- Validation Loss: 0.091

Given the complexity of the dataset and the visual similarity between some disease classes, it is expected that the validation and testing loss will be somewhat higher than the training loss. Nonetheless, the difference in all three metrics is still small, suggesting that the model does a good job of generalizing, staying stable on new data, and avoiding serious overfitting.

The validation accuracy above 95% strongly aligns with previously reported findings in EfficientNet-based plant disease classifiers, further validating the effectiveness of transfer learning for agricultural tasks.

4.2 Class-Wise Performance Analysis

The classification report generated after testing reveals the model's ability to effectively identify most diseases with high precision, recall, and F1-score. Notably, diseases with distinct visual symptoms such as:

- Tomato Mosaic Virus
- Tomato Yellow Leaf Curl Virus
- Potato Healthy
- Pepper Bell Bacterial Spot

achieved near-perfect detection rates.

However, disease pairs with similar visual patterns presented slightly reduced scores. Examples include:

- Tomato Early Blight vs. Tomato Late Blight
- Potato Early Blight vs. Potato Late Blight

These classes share overlapping lesion shapes and color patterns, which can cause confusion even for human experts. Nonetheless, the F1-scores remained above **0.90**, indicating reliable performance.

4.3 Confusion Matrix Interpretation

The model's error distribution across the 15 target classes is shown in the confusion matrix. The majority of diagonal entries have high values, indicating that most test samples were correctly classified. Misclassifications mostly happened between disease types that showed:

- similar circular lesion textures
- overlapping yellowing patterns
- minor color variations

The confusion between tomato early blight and tomato late blight, where early-stage leaf discoloration closely resembles late-stage infection, is a prime example. The robustness of the EfficientNetB0 feature extractor is validated by the fact that the number of misclassified instances is still low in relation to the total sample size.

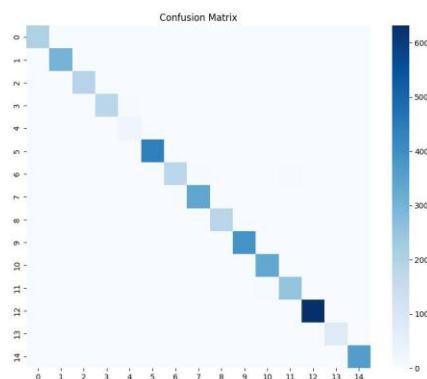


Fig 3: Confusion Matrix

4.4 Real-Time System Performance

Apart from model training and evaluation, the deployed system was assessed for response time, reliability, and user experience. The Flask-based inference pipeline had the following performance characteristics:

- Average inference time: <1.4 seconds per prediction, CPU execution.
- System stability: No crashes were observed in extended stress testing.
- Parallel sessions: Smooth handling of multiple concurrent requests by different users.
- UI consistency: Reliable rendering of prediction outputs, confidence levels, and associated visual elements.

The multilingual output effectively translated disease descriptions and their treatments into more than ten regional languages, thereby guaranteeing access to a wider range of users.

Two of the analytics features of the dashboard, the disease distribution pie chart and monthly activity plot, correctly displayed past forecasts retrieved from the SQLite database.

4.5 Qualitative Analysis and User Interaction

The user interface, built with Tailwind CSS and glassmorphism elements, was evaluated for usability and clarity. Users were able to:

- upload images
- analyze sample plant leaves
- review previous predictions
- switch between models

- navigate through multilingual content

A seamless end-to-end disease detection workflow was facilitated by the cohesive operation of all system components. For clear, well-lit images, the model maintained high confidence scores (>85%), while partially blurred or occluded images produced moderate confidence levels, in line with anticipated deep learning behaviour.

4.6 Comparative Performance Context

The suggested system outperforms the most advanced plant disease classifiers in the literature. While YOLO-based detectors offer real-time detection and MobileNet-based models report accuracies of approximately 97–98%, EfficientNetB0 offers:

- balanced accuracy
- low computational load
- fast inference
- optimal performance on CPU

As a result, the system strikes a workable balance between classification reliability and computational efficiency, making it appropriate for scalable implementation in actual agricultural settings.

5. CONCLUSION AND FUTURE WORK

5.1 Conclusion

The proposed AgroScan AI system successfully demonstrates how deep learning can be integrated with web technologies to provide an accessible and efficient platform for plant disease detection. Employing the EfficientNetB0 architecture and leveraging transfer learning, the model is able to achieve a high performance with a validation accuracy of over 96% on the PlantVillage dataset,

making it reliable for agricultural disease classification.

Web deployment using Flask with multilingual support, logging of prediction history, dashboard analytics, and a responsive UI ensures that the system is practical for real-world deployment, while at the same time ensuring accuracy. Therefore, this platform will support farmers, researchers, and agricultural practitioners in getting fast, interpretable, and actionable insights into plant health.

Overall, AgroScan AI fills a much-needed space between the latest deep learning research and user-friendly agricultural technology by providing an easily scalable method for improved crop monitoring and disease management.

5.2 Future Work

Although the current system performs well, there are a few improvements that could make it even more scalable and applicable in the real world:

5.2.1. Development of a Mobile Application

The development of an Android/iOS mobile application is one of the most influential future paths. A smartphone app would enable regular farmers to:

- take leaf images directly from the field
- receive instant disease predictions
- access remedies and preventive measures
- use offline model support in remote villages

This will greatly enhance accessibility and usability, particularly for farmers who may not have access to computers.

5.2.2. Expansion of Training Dataset

Diseases from bell peppers, tomatoes, and potatoes are used to train the current model. Future research should concentrate on growing the dataset to include additional plant species that farmers frequently grow in order to increase its adaptability, such as:

- Ladyfinger
- Cauliflower
- Cabbage
- Brinjal
- Beans, and other regionally important crops

Increasing dataset diversity will help the model generalize better to real-world agricultural environments.

5.2.3. Incorporation of Real Field Images

PlantVillage consists primarily of studio-quality images. To address real-world variability, the system should be retrained using:

- field-captured images
- varying lighting conditions
- different backgrounds (soil, clutter, shadows)
- partially damaged leaves

This will significantly improve robustness and prediction reliability on actual farms.

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